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SEMICONDUCTOR DEVICE WITH CORNER ISOLATION PROTECTION AND METHODS OF FORMING THE SAME

Abstract

Semiconductor device and the manufacturing method thereof are disclosed. An exemplary semiconductor device comprises a semiconductor stack including semiconductor layers over a substrate, wherein the semiconductor layers are separated from each other and are stacked up along a direction substantially perpendicular to a top surface of the substrate; an isolation structure around a bottom portion of the semiconductor stack and separating active regions; a metal gate structure over a channel region of the semiconductor stack and wrapping each of the semiconductor layers; a gate spacer over a source/drain (S/D) region of the semiconductor stack and along sidewalls of a top portion of the metal gate structure; and an inner spacer over the S/D region of the semiconductor stack and along sidewalls of lower portions of the metal gate structure and wrapping edge portions of each of the semiconductor layers.

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Background/Summary

PRIORITY DATA [0001] This application is a continuation application of U.S. patent application Ser. No. 18/446,665, filed Aug. 9, 2023, which is a divisional application of U.S. patent application Ser. No. 17/213,402, filed Mar. 26, 2021, each of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] The integrated circuit (IC) industry has experienced exponential growth. Multi-gate devices have been introduced to improve gate control by increasing gate-channel coupling and reducing off-state current. One such multi-gate device is a nanosheet device. A nanosheet device substantially refers to any device having a channel region including separated semiconductor channels, and a gate structure, or portions thereof, formed on more than one side of the semiconductor channels (for example, surrounding the semiconductor channels). In some instances, a nanosheet device is also called as a nanowire device, a nanoring device, a gate-surrounding device, a gate-all-around (GAA) device, or a multi-channel bridge device. Nanosheet transistors are compatible with conventional complementary metal-oxide-semiconductor (CMOS) fabrication processes and allow aggressive scaling down of transistors.

[0003] However, fabrication of nanosheet transistors presents challenges. For example, in a conventional nanosheet device, the isolation material between the bottom portion of the semiconductor stack and the metal gate is thin, which may cause high parasitic capacitance between the bottom portion of the metal gate and the semiconductor stack. Therefore, improvement is needed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 illustrates a flowchart of an example method for making an example semiconductor device in accordance with some embodiments of the present disclosure.

[0006] FIG. 2 illustrates a three-dimensional perspective view of the initial example semiconductor device in accordance with some embodiments of the present disclosure.

[0007] FIGS. 3A-13A illustrate top views of the semiconductor device in the three-dimensional perspective view at intermediate stages of the method of FIG. 1 in accordance with some embodiments of the present disclosure.

[0008] FIGS. 3B-13B illustrate cross-sectional views of the semiconductor device along line B-B' in the three-dimensional perspective view at intermediate stages of the method of FIG. 1 in accordance with some embodiments of the present disclosure.

[0009] FIGS. 3C-13C illustrate cross-sectional views of the semiconductor device along line C-C' in the three-dimensional perspective view at intermediate stages of the method of FIG. 1 in accordance with some embodiments of the present disclosure.

[0010] FIGS. 3D-13D illustrate cross-sectional views of the semiconductor device along line D-D' in the three-dimensional perspective view at intermediate stages of the method of FIG. 1 in accordance with some embodiments of the present disclosure.

[0011] FIGS. 10E is an enlarged view of the portion X in FIG. 10C.

DETAILED DESCRIPTION

[0012] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may comprise embodiments in which the first and second features are formed in direct contact, and may also comprise embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact.

[0013] In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Moreover, the formation of a feature on, connected to, and/or coupled to another feature in the present disclosure that follows may comprise embodiments in which the features are formed in direct contact, and may also comprise embodiments in which additional features may be formed interposing the features, such that the features may not be in direct contact. In addition, spatially relative terms, for example, “lower,” “upper,” “horizontal,” “vertical,” “above,” “over,” “below,” “beneath,” “up,” “down,” “top,” “bottom,” etc. as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) are used for ease of the present disclosure of one features relationship to another feature. The spatially relative terms are intended to cover different orientations of the device including the features. Still further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range including the number described, such as within $\pm 10\%$ of the number described or other values as understood by person skilled in the art. For example, the term “about 5 nm” encompasses the dimension range from 4.5 nm to 5.5 nm.

[0014] The present disclosure is generally related to semiconductor devices and the fabrication thereof, and more particularly to methods of fabricating field-effect transistors (FETs), such as nanosheet FETs.

[0015] In a nanosheet device, a channel region of a single device may comprise multiple layers of semiconductor material physically separated from one another. In some examples, a gate of the device is disposed above, alongside, and even between the semiconductor layers of the device. In a conventional nanosheet device, the thickness of the silicon capping layer over the semiconductor stack (for example, to protect the semiconductor stack from being oxidized) is limited, because the silicon capping layer need to be fully oxidized such that it can be fully removed before the channel semiconductor layers are released. When removing the silicon capping layer, the isolation structure may also be partially removed. Therefore, the isolation material between the bottom portions of the semiconductor stack and the metal gate is thin, which may cause high parasitic capacitance

between bottom portions of the metal gate and the semiconductor stack. In the present disclosure, a semiconductor capping layer including the same material as the sacrificial semiconductor layer (for example, including silicon germanium) is formed to protect the isolation structure when removing the dielectric isolation layer. Since the thickness of the semiconductor capping layer can be greater than the conventional silicon capping layer, the isolation material between the bottom portions of the semiconductor stack and the metal gate is thicker. Thus, the parasitic capacitance therebetween can be reduced. In addition, in the present disclosure, the inner spacer around the channel semiconductor layers has a more uniform profile, thus the gate to source/drain (S/D) capacitance and the overlap resistance is more uniform. Thereby, the performance of the semiconductor device is improved.

[0016] FIG. 1 illustrates a flow chart of a method **100** for making an example semiconductor device **200** (hereinafter, device **200**) in accordance with some embodiments of the present disclosure. Method **100** is merely an example and is not intended to limit the present disclosure beyond what is explicitly recited in the claims. Additional operations can be performed before, during, and after method **100**, and some operations described can be replaced, eliminated, or moved around for additional embodiments of the method. Method **100** is described below in conjunction with other figures, which illustrate various three-dimensional and cross-sectional views of the device **200** during intermediate steps of method **100**. In particular, FIG. 2 is a three-dimensional view of an initial structure of device **200** in accordance with some embodiments of the present disclosure. FIGS. 3A-13A illustrate top views of the device **200** shown in FIG. 2 (that is, in an X-Y plane) at intermediate stages of the method **100** in accordance with some embodiments of the present disclosure. FIGS. 3B-13B illustrate cross-sectional views of the device **200** taken along the plane B-B' shown in FIG. 2 (that is, in an X-Z plane) at intermediate stages of the method **100** in accordance with some embodiments of the present disclosure. FIGS. 3C-13C illustrate cross-sectional views of the device **200** taken along the plane C-C' shown in FIG. 2 (that is, in a first Y-Z plane) at intermediate stages of the method **100** in accordance with some embodiments of the present disclosure. FIGS. 3D-13D illustrate cross-sectional views of the device **200** taken along the plane D-D' shown in FIG. 2 (that is, in a second Y-Z plane) at intermediate stages of the method **100** in accordance with some embodiments of the present disclosure.

[0017] Device **200** may be an intermediate device fabricated during processing of an integrated circuit (IC), or a portion thereof, that may comprise static random-access memory (SRAM) and/or other logic circuits, passive components such as resistors, capacitors, and inductors, and active components such as p-type FETs (PFETs), n-type FETs (NFETs), metal-oxide semiconductor field effect transistors (MOSFET), complementary metal-oxide semiconductor (CMOS) transistors, bipolar transistors, high voltage transistors, high frequency transistors, and/or other memory cells. Device **200** can be a portion of a core region (often referred to as a logic region), a memory region (such as a static random access memory (SRAM) region), an analog region, a peripheral region (often referred to as an input/output (I/O) region), a dummy region, other suitable region, or combinations thereof, of an integrated circuit (IC). In some embodiments, device **200** may be a portion of an IC chip, a system on chip (SoC), or portion thereof. The present disclosure is not limited to any particular number of devices or device regions, or to any particular device configurations.

[0018] Referring to FIGS. 1, 2, and 3A-3D, at operation **102**, an initial semiconductor structure of device **200** is formed. As depicted in FIGS. 2 and 3A-3D, device **200** comprises a substrate **202**. In the depicted embodiment, the substrate **202** is a bulk silicon substrate. Alternatively or additionally, the substrate **202** includes another single crystalline semiconductor, such as germanium; a compound semiconductor; an alloy semiconductor; or combinations thereof. Alternatively, the substrate **202** is a semiconductor-on-insulator substrate, such as a silicon-on-insulator (SOI) substrate, a silicon germanium-on-insulator (SGOI) substrate, or a germanium-on-insulator (GOI) substrate. The substrate **202** may be doped with different dopants to form various doped regions

therein. For example, the substrate **202** may include PFET region comprising n-type doped substrate regions (such as n-well) and NFET region comprising p-type doped substrate regions (such as p-well).

[0019] The device **200** includes alternating semiconductor layers formed over the substrate **202**, such as semiconductor layers **210A** including a first semiconductor material and semiconductor layers **210B** including a second semiconductor material that is different from the first semiconductor material. The different semiconductor materials of the semiconductor layers **210A** and **210B** have different oxidation rates and/or different etch selectivity. In some embodiments, the first semiconductor material of the semiconductor layers **210A** is the same as the substrate **202**. For example, the semiconductor layers **210A** comprise silicon (Si, like the substrate **202**), and the semiconductor layers **210B** comprise silicon germanium (SiGe). Thus, alternating SiGe/Si/SiGe/Si/. . . layers are arranged from bottom to top. In some embodiments, the material of the top semiconductor layer may or may not be the same as the bottom semiconductor layer. In some embodiments, no intentional doping is performed when forming the semiconductor layers **210A**. In some other embodiments, the semiconductor layers **210A** may be doped with a p-type dopant or an n-type dopant. The number of the semiconductor layers **210A** and **210B** depends on the design requirements of device **200**. For example, it may comprise one to ten layers of semiconductor layers **210A** or **210B** each. In some embodiments, different semiconductor layers **210A** and **210B** have the same thickness in the Z-direction. In some other embodiments, different semiconductor layers **210A** and **210B** have different thicknesses. For example, each of the first semiconductor layers **210A** has a thickness H1 in the Z-direction, each of the second semiconductor layers **210B** has a thickness H2 in the Z-direction, and the thickness H2 is less than the thickness H1. In some embodiment, the thickness H1 is about 5 nm to about 20 nm and the thickness H2 is about 3 nm to about 15 nm. In some embodiments, the thickness H1 is about 100% to about 200% of the thickness H2. In some embodiments, the semiconductor layers **210A** and/or **210B** are formed by suitable epitaxy process. For example, semiconductor layers comprising SiGe and Si are formed alternately over the substrate **202** by a molecular beam epitaxy (MBE) process, a chemical vapor deposition (CVD) process, such as a metal organic CVD (MOCVD) process, and/or other suitable epitaxial growth processes.

[0020] Thereafter, the alternating semiconductor layers **210A** and **210B** are patterned to form semiconductor stacks **210** (hereinafter the stacks **210**). In some embodiments, various photoresist lithography and etching processes may be performed to the semiconductor layers **210A** and **210B** to form the stacks **210** in fin-shapes as illustrated. For example, first, a patterned photoresist mask is formed over the device **200**. The patterned photoresist mask covers the fin positions according to the design requirement of device **200**. Subsequently, one or more etching processes are performed using the patterned photoresist mask to remove the exposed portions of the first and second semiconductor layers **210A** and **210B**. The remained portions of the first and second semiconductor layers **210A** and **210B** form the fin-shape stacks **210**. In some embodiments, a top portion of the substrate **202** is also patterned to form bottom portions **210C** of the stacks **210**. The etching process includes dry etching, wet etching, other suitable etching process, or combinations thereof. And, the photoresist mask is then removed using any proper method. As depicted in FIG. 3C, a distance between the bottom portions **210C** of the stacks **210** is W1 (in the Y-direction).

[0021] Thereafter, an isolation structure **204** is formed in the trenches between the stacks **210** to separate and isolate the active regions of device **200**. The isolation structure **204** can be configured as different structures, such as a shallow trench isolation (STI) structure, a deep trench isolation (DTI) structure, a local oxidation of silicon (LOCOS) structure, or combinations thereof. The isolation structure **204** includes an isolation material, such as silicon oxide (SiO), silicon nitride (SiN), silicon oxynitride (SiON), other suitable dielectric material (for example, including silicon, oxygen, nitrogen, carbon, and/or other suitable isolation constituent), or combinations thereof. In some embodiments, one or more isolation materials, is deposited over the substrate **202** along

sidewalls of the stacks **210**. The dielectric material may be deposited by CVD (such as plasma enhanced CVD (PECVD)), physical vapor deposition (PVD), thermal oxidation, or other techniques. Subsequently, the dielectric material is recessed (for example, by etching and/or chemical mechanical polishing (CMP)) to form the isolation structure **204**. In some embodiments, a top surface of the isolation structure **204** is substantially coplanar with or below a top surface of the bottom portion **210C** of the stacks **210**. In other words, the isolation structure **204** surrounds the bottom portions **210C** of the stacks **210**. And a width along the Y-direction of the isolation structure **204** between the adjacent stacks **210** is $W1$. Referring to FIGS. 3C and 3D, the isolation structure **204** has a continuous curved (recessed) top surface between the stacks **210**.

[0022] Thereafter, referring to FIGS. 1 and 4A-4D, at operation **104**, a capping layer **212** is formed over the substrate **202**. Referring to FIGS. 4A-4D, the capping layer **212** is formed over the top surfaces and around sidewalls of the stacks **210**. The capping layer **212** is also formed over the isolation structure **204**. In some embodiments, the capping layer **212** includes a material same as the semiconductor layers **210B**. Therefore, in a later process to release the channel semiconductor layers (i.e. semiconductor layers **210A**), the capping layer **212** and the semiconductor layers **210B** can be removed in the same process. In some embodiments, the semiconductor layers **210B** includes SiGe, and the capping layer **212** includes SiGe as well. The capping layer **212** is formed by a deposition process, such as CVD (for example, PECVD), PVD, and/or other suitable deposition process. In some embodiments, after depositing the capping layer **212** (for example, using PECVD), a top surface of the capping layer **212** may be oxidized. For example, if the capping layer **212** includes SiGe, the top surface of the capping layer **212** may include SiGeO.

[0023] Due to the different materials of the stacks **210** (i.e. semiconductor material) and the isolation structure **204** (i.e. dielectric material), the capping layer **212** has different formation speeds around the stacks **210** and over the isolation structure **204**. Referring to FIGS. 4C and 4D, the formation speed of the capping layer **212** around the stacks **210** is faster than that over the isolation structure **204**. Therefore, a bump **212B** is formed at a corner interfacing the stacks **210** and the isolation structure **204**. Referring to FIGS. 4C, the capping layer **212** has a thickness $H3$ in the Z-direction over the stacks **210**. In some embodiments, the thickness $H3$ is about 50% to about 200% of the thickness $H2$ of the semiconductor layers **210B**. In some further embodiments, the thickness $H3$ is about 3 nm to about 10 nm. If the thickness $H3$ is more than 200% of the $H2$ or if the thickness $H3$ is more than 10 nm, the capping layer **212** may merged between the stacks **210**, thus left no space for the gate structures and spacers to be formed between the stacks. On the other hand, if the thickness $H3$ is less than 50% of the $H2$ or if the thickness $H3$ is less than 3 nm, the bump **212B** may not be big enough to protect the corner isolation material during the dummy gate structure and isolation layer **214** (FIGS. 5A-5D) removing process, the parasitic capacitance may be increased in some instances. The capping layer **212** has a thickness $W2$ in the Y-direction along the sidewalls of the stacks **210**. In some embodiments, the thickness $W2$ is 10%~30% of the distance $W1$ between the stacks **210**. The bump **212B** of the capping layer **212** has a thickness $W3$ in the Y-direction. In some embodiments, the thickness $W3$ is about 120% to about 200% of the thickness $W2$. In some further embodiments, the thickness $W3$ is about 2 nm to about 15 nm. Similarly, the thickness $W3$ of the bump **212B** cannot be too big or too small. If $W3$ is too big, the adjacent stacks **210** may be connected, thus left no space for the gate structure and gate spacers to be formed therebetween; or, if $W3$ is too small, it cannot efficiently protect the corner isolation material and the parasitic capacitance may be increased.

[0024] Referring to FIGS. 1 and 5A-5D, at operation **106**, an isolation layer **214** is formed on the capping layer **212**, i.e. over the top surfaces and around sidewalls of the stacks **210**, and over the isolation structure **204**. In some embodiments, the isolation layer **214** is formed to protect the stacks **210** from being oxidized. In some embodiments, the isolation layer **214** includes a dielectric material such as SiO, SiN, SiON, other suitable dielectric material, or combinations thereof. The dielectric isolation layer **214** is formed by a deposition process, such as atomic layer deposition

(ALD), CVD, PVD, and/or other suitable deposition process.

[0025] Referring to FIGS. **1** and **6A-6D**, at operation **108**, dummy gate structures **220** are formed over the stacks **210**. Each dummy gate structure **220** serves as a placeholder for subsequently forming a metal gate structure. In some embodiments, the dummy gate structures **220** extend along the Y-direction and traverse respective stacks **210**. The dummy gate structures **220** cover the channel regions of the stacks **210** which interpose the source regions and the drain regions (both referred to as the S/D regions). Each of the dummy gate structures **220** may include various dummy layers, for example, an interfacial layer (including a dielectric material such as SiN, SiO, SiON, etc.), a dummy gate electrode (including polysilicon), one or more hard mask layers (including a dielectric material such as SiN, silicon carbonitride (SiCN), SiO, etc.), and/or other suitable layers. The dummy gate structures **220** are formed by deposition processes, lithography processes, etching processes, other suitable processes, or combinations thereof. For example, first, different dummy gate layers are deposited over the stacks **210**. A lithography process is then performed to form a mask covering the channel regions of the stacks **210**. Thereafter, the different dummy gate layers are etched using the lithography mask to form the dummy gate structures **220**. In some embodiments, referring to FIG. **6B**, portions of the isolation layer **214** uncovered by the lithography mask are also removed by the etching process. The etching process may be a dry etch, a wet etch, or combinations thereof. Referring to FIG. **6C**, the isolation layer **214** covered by the dummy gate structures **220** is remained. And, the lithography mask is then removed using any proper method.

[0026] Still referring to FIGS. **1** and **6A-6D**, at operation **108**, gate spacers **222** are formed along sidewalls of the dummy gate structures **220** in the S/D regions of the stacks **210**. In some embodiments, the gate spacers **222** comprises a dielectric material, such as SiO, SiN, SiON, silicon carbide (SiC), other dielectric material, or a combination thereof. In some embodiments, the gate spacers **222** are single layer structures. In some other embodiments, the gate spacers **222** includes multiple layers. The gate spacers **222** are formed by various deposition and etching processes. For example, the dielectric material of the gate spacers **222** are deposited over the substrate **202** by a deposition process (for example, by ALD, CVD, PVD, or other proper process) to form a gate spacer layer. The gate spacer layer is formed over the isolation structure **204**, along sidewalls and over top surfaces of the dummy gate structures **220**, and along sidewalls and over top surfaces of the stacks **210**. Thereafter, an anisotropically etching process is performed to remove the portions of the spacer layer in the X-Y plane (the plane in which the top surface of the substrate **202** is). The remaining portions of the spacer layer along sidewalls of the dummy gate structures **220** and around the stacks **210** become the gate spacers **222**. The anisotropically etching includes wet etching, dry etching, or combinations thereof. As depicted in FIG. **6D**, the gate spacers **222** covers and surrounds the isolation layer **212** in the S/D region.

[0027] Referring to FIGS. **1** and **7A-7D**, at operation **110**, S/D trenches **224** are formed in the S/D regions of the stacks **210**. In some embodiments, the stacks **210** are recessed by a S/D etching process along sidewalls of the gate spacers **222** to form the S/D trenches **224**. The S/D etching process may be a dry etching process (such as a reactive ion etching (RIE) process), a wet etching process, or combinations thereof. The duration of the S/D etching process is controlled such that the sidewalls of each semiconductor layers **210A** and **210B**, as well as the bottom portion **210C**, are exposed in the S/D trenches **224**.

[0028] Now referring to FIGS. **1**, **8A-8D**, at operation **112**, inner spacers **230** are formed around the edge portions of the semiconductor layers **210A**. In some embodiments, first, the portions (edges) of the semiconductor layers **210B** and the isolation layer **212** exposed in the S/D trenches **224** are selectively removed by a suitable etching process to form gaps between the edge portions of the semiconductor layers **210A**. In other words, the edge portions of the semiconductor layers **210A** are suspended in the S/D trenches **224**. Due to the different oxidation rates and/or etching selectivities of the materials of the semiconductor layers **210A** (for example, Si) and the

semiconductor layers **210B** or the capping layer **212** (for example, SiGe), only exposed portions (edges) of the semiconductor layers **210B** and the capping layer **212** are removed, while the semiconductor layers **210A** remain substantially unchanged. In some embodiments, the selective removal of the exposed portions of the semiconductor layers **210B** may include an oxidation process followed by a selective etching process. For example, the edge portions of the semiconductor layers **210B** and the capping layer **212** are first selectively oxidized to include a material of SiGeO. Then, a selective etching process is performed to remove the SiGeO with a suitable etchant such as ammonium hydroxide (NH₄OH) or hydro fluoride (HF). The duration of the oxidation process and the selective etching process can be controlled such that only edge portions of the semiconductor layers **210B** and the capping layer **212** are selectively removed.

[0029] Thereafter, inner spacers **230** are formed surrounding the edge portions of the semiconductor layers **210A**. The inner spacers **230** comprise a dielectric material which can be same as or different from the material of the gate spacers **222**. In some embodiments, the inner spacers **230** comprise a dielectric material such as SiO, SiN, SiON, SiC, or combinations thereof. The dielectric material of the inner spacers may be deposited in the S/D trenches **224** and fill in the gaps around the edge portions of the semiconductor layers **210A** by CVD, PVD, ALD, or combinations thereof. Extra dielectric material is then removed along sidewalls of the gate spacers **222** until the sidewalls of the semiconductor layers **210A** are exposed in the S/D trenches **224**.

[0030] In a conventional nanosheet device where no SiGe capping layer formed around the semiconductor stacks, the inner spacers only fill in the gaps between the edge portions of the channel semiconductor layers (for example, the semiconductor layers **210A**). However, in the present disclosure, due to the SiGe capping layer, the inner spacers **230** not only include the portions **230A** fill in the gaps between the edge portions of the channel semiconductor layers (i.e. semiconductor layers **210A**), but also a portion **230T** above the topmost channel semiconductor layer, and side portions **230S** along sidewalls of the channel semiconductor layers. The inner spacers also include bump portions **230B** at the corner between the isolation structure **204** and the semiconductor stacks **210**. Therefore, the inner spacer has a more uniform shape around the channel semiconductor layers than the conventional nanosheet device, which may result in a more uniform gate to S/D capacitance and a more uniform overlay resistance.

[0031] Referring to FIG. **8D**, the horizontal portions **230A** of the inner spacer has a thickness H₂, and the horizontal portion **230T** of the inner spacer has a thickness H₃, which is about 50% to about 200% of the thickness H₂. The vertical side portions **230S** has a width W₂ which is less than about 10% of the width W₁ of the isolation structure **204** between the stacks **210**. The bump portion **230B** has a width W₃, which is about 120% to about 200% of W₂. The various sizes of the inner spacer **230** are comply with the sizes of the capping layer **212**.

[0032] Now referring to FIGS. **1** and **9A-9D**, at operation **114**, the S/D features **240** are epitaxially grown in the S/D trenches **224**. In some embodiments, the epitaxial S/D features **240** include a semiconductor material such as Si or Ge; a compound semiconductor such as SiGe, SiC, gallium arsenide (GaAs), etc.; an alloy semiconductor; or combinations thereof. An epitaxy process may be implemented to epitaxially grow the S/D features **240**. The epitaxy process may comprise CVD deposition (for example, vapor-phase epitaxy (VPE), ultra-high vacuum CVD (UHV-CVD), low-pressure CVD (LPCVD), and/or plasma-enhanced (PECVD)), molecular beam epitaxy (MBE), other suitable selective epitaxial growth (SEG) processes, or combinations thereof. In some embodiments, the epitaxial S/D features **240** are separated for different stacks **210** and are not merged in the Y-direction, as depicted in FIG. **9A**. In some other embodiments, the epitaxial S/D features **240** may be merged in the Y-direction and crossing more than one stacks **210**.

[0033] Now referring to FIGS. **1** and **10A-10D**, at operation **116**, the dummy gate structures **220** and the isolation layer **214** are removed to be prepared for a metal gate replacement process. FIGS. **10E** is an enlarged view of the portion X in FIG. **10C**. Referring to FIGS. **10A** and **10B**, an interlayer dielectric (ILD) layer **242** is formed over the substrate **202**. The ILD layer **242** is

disposed along the gate spacers **222**, over the epitaxial S/D features **240** and the isolation structure **204** in the S/D regions. In some embodiments, the ILD layer **242** comprises a low-k ($K < 3.9$) dielectric material, such as tetraethylorthosilicate (TEOS), un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fluorosilicate glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), other suitable dielectric materials, or combinations thereof. The ILD layer **242** may be formed by deposition processes such as CVD, flowable CVD (FCVD), spin-on-glass (SOG), other suitable methods, or combinations thereof.

[0034] The dummy gate structures **220** and the isolation layer **214** are then removed to form gate trenches **244** (referring to FIG. **10B**) exposing the channel regions of the stacks **210**. In some embodiments, removing the dummy gate structures **220** and the isolation layer **214** comprises one or more etching processes, such as wet etching, dry etching, or other etching techniques. The capping layer **212** and the isolation structure **204** are then exposed in the gate trenches **244**. Referring to FIG. **10C**, the removal of the isolation layer **214** at operation **116** also removes portions of the isolation structure **204** (uncovered by the capping layer **212**) between the stacks **210** in the channel region. And, referring to FIG. **10D**, portions of the isolation structure **204** (covered by the gate spacer **222** and the inner spacers **230**) in the S/D regions are substantially unchanged. Therefore, the portions of the isolation structure **204** in the channel region and the portions of the isolation structure **204** in the S/D region have different shapes. As depicted in FIG. **10E**, the uncovered portion of the isolation structure **204** in the channel region can be recessed to a depth **H4**, for example, about 2 nm to about 10 nm, due to the etching process limitation. The isolation structure **204** in the channel region includes an upper portion **204U** (covered by the capping layer **212**) and a lower portion **204L** (uncovered by the capping layer **212**). Sidewalls of the upper portion **204U** and the lower portion **204L** are discontinuously connected. In other words, sidewalls of the upper portion **204U** has a gradient **G1**, and sidewalls of the lower portion **204L** has a gradient **G2** greater than **G1**, referring to the X-Y plane. As depicted in FIG. **10D**, the isolation structure **204** in the S/D region (covered by the gate spacer **222** and the inner spacers **230**) has a continuous curved (recessed) top surface. In some embodiments, a portion of the capping layer **212** (for example, less than about 20%) may also be trimmed when removing the isolation layer **214**.

[0035] Referring to FIGS. **1**, **11A-11D** and **12A-12D**, at operation **118**, a metal gate replacement process is performed. The metal gate replacement process includes various processes. Referring to FIGS. **11A-11D**, the capping layer **212** and the semiconductor layers **210B** are selectively removed from the gate trenches **244**. Due to the different materials of the semiconductor layers **210A** (for example, Si) and the semiconductor layers **210B** or the capping layer **212** (for example, SiGe), the semiconductor layers **210B** and the capping layer **212** are removed by a selective oxidation/etching process similar as those to remove the edge portions of the semiconductor layers **210B** and the capping layer **212** during the inner spacer formation process. In some embodiments, the semiconductor layers **210A** are slightly etched or not etched. Thereby, the semiconductor layers **210A** are suspended in the channel regions and stacked up along the direction (Z-direction) substantially perpendicular to the top surface of the substrate **202** (X-Y plane). The suspended semiconductor layers **210A** are also referred to as channel semiconductor layers **210A**.

[0036] Then, referring to FIGS. **12A-12D**, metal gate structures **250** are formed in the channel regions of the stacks **210**. The metal gate structures **250** wrap each of the suspended semiconductor layers **210A**. In some embodiments, each metal gate structure **250** may include a gate dielectric layer wrapping around each of the channel semiconductor layers **210A**, a metal gate electrode over the gate dielectric layer, and other suitable layers. The gate dielectric layer includes a high-k ($K > 3.9$) dielectric material, such as HfO₂, HfSiO, HfSiO₄, HfSiON, HfLaO, HfTaO, HfTiO, HfZrO, HfAlO_x, ZrO, ZrO₂, ZrSiO₂, AlO, AlSiO, Al₂O₃, TiO, TiO₂, LaO, LaSiO, Ta₂O₃, Ta₂O₅, Y₂O₃, SrTiO₃, BaZrO, BaTiO₃ (BTO), (Ba,Sr) TiO₃ (BST), Si₃N₄, hafnium dioxide-alumina (HfO₂—Al₂O₃) alloy, other suitable high-k dielectric material, or combinations thereof. In some embodiments, the gate dielectric layer is deposited by CVD, PVD, ALD, and/or other suitable

method. In some embodiments, each metal gate electrode includes one or more work function metal (WFM) layers and a bulk metal. The WFM layer is configured to tune a work function of its corresponding transistor to achieve a desired threshold voltage V_t . And, the bulk metal is configured to serve as the main conductive portion of the functional gate structure. In some embodiments, the material of the WFM layer may include TiAl, TiAlC, TaAlC, TiAlN, TiN, TSN, TaN, WCN, Mo, other materials, or combinations thereof. The bulk metal may include Al, W, Cu, or combinations thereof. The various layers of the metal gate electrode may be formed by any suitable method, such as CVD, ALD, PVD, plating, chemical oxidation, thermal oxidation, other suitable methods, or combinations thereof. Thereafter, one or more polishing processes (for example, CMP) are applied to remove any excess conductive materials and planarize the top surface of the device **200**.

[0037] In a conventional nanosheet device, the isolation structure around the bottom portion of the semiconductor stacks is trimmed to be very thin during the silicon capping layer removing process. Therefore, after the formation of the metal gate, the parasitic capacitance between the metal gate and the bottom portion of the semiconductor stack is high. However, in the present disclosure, due to the protection of the capping layer (having the same semiconductor material as the sacrificial semiconductor layers **210B**), the isolation structure around the bottom portion of the semiconductor stack is thicker, thereby the parasitic capacitance between the metal gate and the bottom portion of the semiconductor stack can be reduced. Therefore, the performance of the nanosheet device can be improved.

[0038] Now referring to FIGS. **1**, **13A-13D**, at operation **120**, further processing is performed to complete the fabrication of the device **200**. For example, it may form various contacts/vias **270**, metal lines (not shown), as well as other multilayer interconnect features, such as etch stop layer (ESL **260**) and/or ILD layers **262** over the device **200**, configured to connect the various features to form a functional circuit that comprises the different semiconductor devices.

[0039] Although not intended to be limiting, one or more embodiments of the present disclosure provide many benefits to a semiconductor device and a formation process thereof. For example, embodiments of the present disclosure provide a semiconductor device including corner isolation protection to enlarge the thickness of the isolation structure around the bottom portions of the semiconductor device. The thicker isolation structure can reduce the capacitance between the bottom portions of the metal gate and the semiconductor stack. The semiconductor device of the present disclosure also includes a uniform inner spacer around edge portions of each of the channel semiconductor layers of the semiconductor stack, which can provide more uniform gate to S/D capacitance and more uniform overlay resistance compare with the conventional structure. Therefore, the performance of the semiconductor device is improved.

[0040] The present disclosure provides for many different embodiments. Semiconductor device having corner isolation protection and methods of fabrication thereof are disclosed herein. An exemplary semiconductor device comprises a semiconductor stack including semiconductor layers over a substrate, wherein the semiconductor layers are separated from each other and are stacked up along a direction substantially perpendicular to a top surface of the substrate; an isolation structure around a bottom portion of the semiconductor stack and separating active regions; a metal gate structure over a channel region of the semiconductor stack and wrapping each of the semiconductor layers; a gate spacer over a source/drain (S/D) region of the semiconductor stack and along sidewalls of a top portion of the metal gate structure; and an inner spacer over the S/D region of the semiconductor stack and along sidewalls of lower portions of the metal gate structure and wrapping edge portions of each of the semiconductor layers.

[0041] In some embodiments, the isolation structure has a first shape in the channel region and a second shape in the S/D region, the second shape is different from the first shape. In some further embodiments, the first shape of the isolation structure in the channel region includes an upper portion and a lower portion, wherein a gradient of sidewalls of the upper portion is less than a

gradient of sidewalls of the lower portion. In some further embodiments, the second shape of the isolation structure in the S/D region is a continuous recessing shape.

[0042] In some embodiments, a side portion of the inner spacer includes a bump contacting the isolation structure.

[0043] In some embodiments, the inner spacer covers a top surface of a topmost semiconductor layer and contacting a sidewall of each semiconductor layer in an edge portion. In some further embodiments, the inner spacer has a first thickness over the top surface of the top most semiconductor layer and a second thickness between the semiconductor layers, and the first thickness is about 50% to about 200% of the second thickness. In some further embodiments, the gate spacer is over the inner spacer and surrounds sidewalls of the inner spacer.

[0044] Another exemplary semiconductor device comprises semiconductor stacks over a substrate, wherein each semiconductor stack includes semiconductor layers separated from each other and stacked up along a direction substantially perpendicular to a top surface of the substrate; a metal gate structure over channel regions of the semiconductor stacks and wrapping each of the semiconductor layers; and an isolation structure separating bottom portions of the semiconductor stacks, wherein the isolation structure includes a first shape in the channel regions and a second shape in source/drain (S/D) regions, wherein the first shape is different from the second shape.

[0045] In some embodiments, the exemplary semiconductor device further comprises inner spacers wrapping edge portions of the semiconductor layers, wherein the inner spacers include horizontal portions wrapping top and bottom surfaces of the semiconductor layers and vertical portions along sidewalls of the semiconductor layers. In some embodiments, the exemplary semiconductor device further comprises a width of the vertical portions of the inner spacers is less than about 10% of a width of the isolation structure between the semiconductor stacks. In some embodiments, the vertical portions of the inner spacers include bumps contacting the isolation structure.

[0046] A method of forming a semiconductor device comprises alternately forming a semiconductor stack including first semiconductor layers and second semiconductor layers over a substrate, wherein the first semiconductor layers and the second semiconductor layers include different materials and are stacked up along a direction substantially perpendicular to a top surface of the substrate; forming an isolation structure around a bottom portion of the semiconductor stack; depositing a third semiconductor layer over the semiconductor stack and the isolation structure, wherein the third semiconductor layer includes a material same as the second semiconductor layers; forming a dummy gate structure over a channel region of the semiconductor stack; and forming a gate spacer along sidewalls of the dummy gate structure, wherein the gate spacer is over and surrounds the third semiconductor layer.

[0047] In some embodiments, the exemplary method further comprises forming a source/drain (S/D) trench in a S/D region of the semiconductor stack; removing edge portions of the second semiconductor layers and the third semiconductor layer from the S/D trench; and forming an inner spacer in the removed edge portions of the second semiconductor layers and the third semiconductor layer, wherein the inner spacer wrapping around the edge portions of each of the first semiconductor layers. In some embodiments, the exemplary method further comprises epitaxially growing a S/D feature in the S/D region, wherein the S/D feature contacting the inner spacer.

[0048] In some embodiments, the exemplary method further comprises removing the dummy gate structure; selectively removing the second semiconductor layers and the third semiconductor layer; and forming a metal gate structure wrapping each of the first semiconductor layers. In some embodiments, the exemplary method further comprises forming a dielectric isolation layer over the third semiconductor layer in the channel region before forming the dummy gate structure. In some embodiments, the exemplary method further comprises removing the dielectric isolation layer and a portion of the isolation structure in the channel region, such that the isolation structure in the channel region includes a first portion and a second portion, and a gradient of sidewalls of the first

portion is less than a gradient of sidewalls of the second portion.

[0049] In some embodiments, the depositing the third semiconductor layer over the semiconductor stack and the isolation structure includes depositing the third semiconductor layer over the semiconductor stack with a higher speed than over the isolation structure. In some further embodiments, the third semiconductor layer includes a bump portion contacting the isolation structure.

[0050] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a substrate; a bottom fin disposed over the substrate; a plurality of channel layers disposed over the bottom fin; a gate structure wrapping around each of the plurality of channel layers; an isolation structure disposed over the substrate and interfacing sidewalls of the bottom fin; a gate spacer disposed along sidewalls of a top portion of the gate structure over the plurality of channel layers; and an interlayer dielectric (ILD) layer adjacent the gate structure and disposed over the isolation structure, wherein a profile of a first portion of the isolation structure disposed below the gate spacer is different from a profile of a second portion of the isolation structure disposed below the ILD layer, wherein the ILD layer and the gate spacer interface the isolation structure.
2. The semiconductor device of claim 1, further comprising: an inner spacer wrapping around an edge portion of each of the plurality of channel layers.
3. The semiconductor device of claim 2, wherein the gate spacer wraps over the inner spacer.
4. The semiconductor device of claim 2, further comprising: a source/drain feature interfacing end sidewalls of the plurality of channel layers.
5. The semiconductor device of claim 4, wherein the gate structure is spaced apart from the source/drain feature by the inner spacer.
6. The semiconductor device of claim 1, wherein top surfaces of the gate structure and the ILD layer, the gate spacer and the gate structure are coplanar.
7. The semiconductor device of claim 6, further comprising: an etch stop layer disposed over and interfacing the top surfaces of the gate structure and the ILD layer, the gate spacer and the gate structure are coplanar.
8. The semiconductor device of claim 1, wherein the profile of the second portion of the isolation structure below the ILD layer comprises a recess, wherein the recess comprises a depth between about 2 nm and about 10 nm.
9. A semiconductor structure, comprising: a substrate; a first active region and a second active region extending lengthwise along a first direction over the substrate, wherein the first active region comprises a first bottom fin and a first plurality of nanostructures disposed over the first bottom fin, wherein the second active region comprises a second bottom fin and a second plurality of nanostructures disposed over the second bottom fin; a first metal gate structure wrapping over each of the first plurality of nanostructures; a second metal gate structure wrapping over each of the second plurality of nanostructures; an isolation structure disposed between and interfacing the first bottom fin and the second bottom fin; a gate spacer disposed along sidewalls of the first metal gate structure and the second metal gate structure; and an interlayer dielectric (ILD) layer disposed

- between the first metal gate structure and the second metal gate structure along a second direction perpendicular to the first direction, wherein a profile of a first portion of the isolation structure disposed below the gate spacer is different from a profile of a second portion of the isolation structure disposed below the ILD layer.
- 10.** The semiconductor structure of claim 9, wherein the profile of the second portion of the isolation structure below the ILD layer comprises a recess.
- 11.** The semiconductor structure of claim 10, wherein the recess comprises a depth between about 2 nm and about 10 nm.
- 12.** The semiconductor structure of claim 10, further comprising: an inner spacer wrapping around an edge portion of each of the first plurality of nanostructures and each of the second plurality of nanostructures.
- 13.** The semiconductor structure of claim 12, wherein the gate spacer wraps over the inner spacer.
- 14.** The semiconductor structure of claim 12, wherein the first bottom fin and the second bottom fin are spaced apart by a distance along a second direction perpendicular to the first direction, wherein the inner spacer comprises a thickness along the second direction, wherein the thickness is between about 10% and about 30% of the distance.
- 15.** The semiconductor structure of claim 12, wherein the inner spacer comprises a bump over the isolation structure.
- 16.** The semiconductor structure of claim 12, wherein the first plurality of nanostructures and the second plurality of nanostructures are spaced apart from the gate spacer by the inner spacer.
- 17.** A semiconductor structure, comprising: a substrate; a first active region and a second active region extending lengthwise along a first direction over the substrate, wherein the first active region comprises a first bottom fin and a first plurality of nanostructures disposed over the first bottom fin, wherein the second active region comprises a second bottom fin and a second plurality of nanostructures disposed over the second bottom fin; an isolation structure disposed between and interfacing the first bottom fin and the second bottom fin; a first metal gate structure wrapping over each of the first plurality of nanostructures; a second metal gate structure wrapping over each of the second plurality of nanostructures; an inner spacer wrapping around an edge portion of each of the first plurality of nanostructures and each of the second plurality of nanostructures; a gate spacer wrapping over the inner spacer; and an interlayer dielectric (ILD) layer disposed between the first metal gate structure and the second metal gate structure along a second direction perpendicular to the first direction, wherein a profile of a first portion of the isolation structure disposed below the gate spacer is different from a profile of a second portion of the isolation structure disposed below the ILD layer.
- 18.** The semiconductor structure of claim 17, wherein the inner spacer and the gate spacer interface the isolation structure.
- 19.** The semiconductor structure of claim 17, wherein the first bottom fin and the second bottom fin are spaced apart by a distance along a second direction perpendicular to the first direction, wherein the inner spacer comprises a thickness along the second direction, wherein the thickness is between about 10% and about 30% of the distance.
- 20.** The semiconductor structure of claim 17, wherein the profile of the second portion of the isolation structure below the ILD layer comprises a recess, wherein the recess comprises a depth between about 2 nm and about 10 nm.
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